Supporting Information

Topologically Exact 3D/2D Line and Ray Clipping for Distributed CT and MRI Volume Computing: The Analytical Phase-Compensated Bresenham Algorithm

Hemanth Manchabale Papachappa[1]

Independent Researcher, Seattle, Washington, USA

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S1 Extended Mathematical Proofs

S1.1 Proof of Lemma 1 (Closed-Form Discrete Coordinate Dynamics)

Let a 3D digital ray have integer endpoints $\mathbf{P}_0 = (x_0, y_0, z_0)$ and $\mathbf{P}_1 = (x_1, y_1, z_1)$ with $\Delta x \geq \Delta y, \Delta z \geq 0$ in canonical Octant 1. The midpoint formulation of the discrete line evaluates at step $k = x - x_0 \in [0, \Delta x]$:

$$y(k) = y_0 + \sum_{j=0}^{k-1} \mathbb{I}(D_{1,j} > 0). \tag{S1}$$

The continuous ray satisfies $y_{\text{ideal}}(k) = y_0 + \frac{\Delta y}{\Delta x} k$. The discrete decision condition at step j checks whether the continuous ray strictly crosses the candidate voxel midpoint $y_j + \frac{1}{2}$:

$$y_0 + \frac{\Delta y}{\Delta x} j > y_j + \frac{1}{2} \iff 2\Delta y \cdot j - 2\Delta x(y_j - y_0) - \Delta x > 0. \tag{S2}$$

By linear induction, $y_j - y_0$ increments at exactly those indices where the ideal line exceeds the midpoint. In integer arithmetic over $\mathbb{Z}$, the strict inequality $u > v + 1/2$ is equivalent to $u \geq v + 1/2 + \epsilon$,

[1]Corresponding Author. E-mail: hemathmpkumar123@gmail.com; ORCID: 0009-0007-1398-9451

which transforms under integer floor division to:

$$y(k) = y_0 + \left\lfloor \frac{2\Delta y \cdot k + \Delta x - 1}{2\Delta x} \right\rfloor. \tag{S3}$$

Subtracting 1 in the numerator guarantees that exact mathematical ties ($y(x) = y + 1/2$) evaluate to the lower integer, replicating standard Bresenham tie-breaking behavior for all $k \in \mathbb{N}$. An identical construction applies to $z(k)$:

$$z(k) = z_0 + \left\lfloor \frac{2\Delta z \cdot k + \Delta x - 1}{2\Delta x} \right\rfloor. \tag{S4}$$

□

S1.2 Proof of Lemma 2 (Closed-Form Accumulated Error States)

The initial Bresenham error state at $k = 0$ is $D_{1,0} = 2\Delta y - \Delta x$. At each step $j \to j + 1$, D_1 undergoes the transition:

$$D_{1,j+1} = D_{1,j} + 2\Delta y - 2\Delta x \cdot \Delta y_j, \tag{S5}$$

where $\Delta y_j = y_{j+1} - y_j \in \{0, 1\}$. Summing this telescoping recurrence from $j = 0$ to $k - 1$:

$$D_1(k) = D_{1,0} + \sum_{j=0}^{k-1} 2\Delta y - \sum_{j=0}^{k-1} 2\Delta x \cdot \Delta y_j = (2\Delta y - \Delta x) + 2\Delta y \cdot k - 2\Delta x(y(k) - y_0). \tag{S6}$$

Because integer additions commute and associate in $\mathbb{Z}$, this closed-form expression evaluates the exact register state directly at any arbitrary step index k in $O(1)$ operations, without evaluating any intermediate steps. By symmetric argument, the dual accumulator $D_2(k)$ satisfies:

$$D_2(k) = (2\Delta z - \Delta x) + 2\Delta z \cdot k - 2\Delta x(z(k) - z_0). \tag{S7}$$

□

S2 Complete Algorithmic Implementation Details

Algorithm S1: 3D AP-Bresenham Sub-Volume Ray Traversal

Input: Global line endpoints $\mathbf{P}_0, \mathbf{P}_1 \in \mathbb{Z}^3$; sub-volume bounds $[X_{\min}, X_{\max}] \times [Y_{\min}, Y_{\max}] \times [Z_{\min}, Z_{\max}]$.

Output: Ordered sequence of traversed 26-connected voxels $\mathcal{V} \subset \mathbb{Z}^3$.

Procedure:

1. Determine major axis and signs:
   - $\Delta x = |x_1 - x_0|, \Delta y = |y_1 - y_0|, \Delta z = |z_1 - z_0|$.
   - Identify primary driving axis $\Delta_{\text{maj}} = \max(\Delta x, \Delta y, \Delta z)$ and secondary axes $\Delta_{\text{min1}}, \Delta_{\text{min2}}$.
   - Map local bounding box to canonical relative bounds $w_{m,\min}, w_{m,\max}$.
2. Compute Diophantine entry/exit bounds along each axis in $O(1)$:

$$k_{\text{maj,min}} = \max(0, w_{\text{maj,min}}), \quad k_{\text{maj,max}} = \min(\Delta_{\text{maj}}, w_{\text{maj,max}})$$

$$k_{1,\min} = \left\lceil \frac{2\Delta_{\text{maj}} \cdot w_{1,\min} - \Delta_{\text{maj}} + 1}{2\Delta_{\text{min1}}} \right\rceil, \quad k_{1,\max} = \left\lfloor \frac{2\Delta_{\text{maj}} \cdot (w_{1,\max} + 1) - \Delta_{\text{maj}}}{2\Delta_{\text{min1}}} \right\rfloor$$

$$k_{2,\min} = \left\lceil \frac{2\Delta_{\text{maj}} \cdot w_{2,\min} - \Delta_{\text{maj}} + 1}{2\Delta_{\text{min2}}} \right\rceil, \quad k_{2,\max} = \left\lfloor \frac{2\Delta_{\text{maj}} \cdot (w_{2,\max} + 1) - \Delta_{\text{maj}}}{2\Delta_{\text{min2}}} \right\rfloor$$

3. Intersect intervals:

$$k_{\text{start}} = \max(k_{\text{maj,min}}, k_{1,\min}, k_{2,\min}), \quad k_{\text{end}} = \min(k_{\text{maj,max}}, k_{1,\max}, k_{2,\max})$$

   if $k_{\text{start}} > k_{\text{end}}$ **then return** $\emptyset$ (ray misses sub-volume).

4. Analytically inject entry voxel and decision accumulators:

$$m_{1,\text{rel}} = \left\lfloor \frac{2\Delta_{\text{min1}} \cdot k_{\text{start}} + \Delta_{\text{maj}} - 1}{2\Delta_{\text{maj}}} \right\rfloor, \quad m_{2,\text{rel}} = \left\lfloor \frac{2\Delta_{\text{min2}} \cdot k_{\text{start}} + \Delta_{\text{maj}} - 1}{2\Delta_{\text{maj}}} \right\rfloor$$

$$D_{1,\text{start}} = (2\Delta_{\text{min1}} - \Delta_{\text{maj}}) + 2\Delta_{\text{min1}} \cdot k_{\text{start}} - 2\Delta_{\text{maj}} \cdot m_{1,\text{rel}}$$

$$D_{2,\text{start}} = (2\Delta_{\text{min2}} - \Delta_{\text{maj}}) + 2\Delta_{\text{min2}} \cdot k_{\text{start}} - 2\Delta_{\text{maj}} \cdot m_{2,\text{rel}}$$

5. Step discrete walker from $k = k_{\text{start}}$ to k_{end} using standard integer increments $(+2\Delta_{\min}, -2\Delta_{\text{maj}})$, emitting un-mapped world voxel coordinates at each step.

Figure S1: Pseudocode for 3D Analytical Phase-Compensated Bresenham (AP-Bresenham).

S3 Extended Ablation Benchmarks across Partition Scales

Table S1 provides detailed scaling metrics across 8, 27, 64, 125, 216, 343, and 512 distributed sub-volumes, comparing conventional slab-clipping and AP-Bresenham on the 3D Shepp-Logan CT head phantom ($N = 10,000$ multi-angle rays).

Table S1: **Extended Multi-Scale Scaling Ablation Study ($N = 10,000$ multi-angle projection rays).**

Sub-Volumes	**Grid Partition**	**Slab-Clip + Reseed (Conventional)**		**3D AP-Bresenham (Ours)**		**Setup Speedup**
		Seam Defect Rate	**CT RMSE / PSNR**	**Seam Defect Rate**	**CT RMSE / PSNR**	
8 chunks	$2 \times 2 \times 2$	99.72%	7.39 / 16.6 dB	**0.00%**	**0.00** / ∞	1.20×
27 chunks	$3 \times 3 \times 3$	99.81%	7.44 / 16.5 dB	**0.00%**	**0.00** / ∞	1.22×
64 chunks	$4 \times 4 \times 4$	99.88%	7.50 / 16.5 dB	**0.00%**	**0.00** / ∞	1.21×
125 chunks	$5 \times 5 \times 5$	99.85%	8.12 / 15.8 dB	**0.00%**	**0.00** / ∞	1.23×
216 chunks	$6 \times 6 \times 6$	99.86%	8.41 / 15.5 dB	**0.00%**	**0.00** / ∞	1.24×
343 chunks	$7 \times 7 \times 7$	99.84%	8.65 / 15.2 dB	**0.00%**	**0.00** / ∞	1.25×
512 chunks	$8 \times 8 \times 8$	99.84%	8.90 / 15.0 dB	**0.00%**	**0.00** / ∞	1.25×